DPEN022: Enabling Mathematics B
Class Questions Week 3

<u>Week 3A</u>

Key Idea – Derivative of Powers of *x* and using Constant Rule

1. Differentiate the following, writing your answer in positive index form: *[power is an integer]*

(a) $f(x)=7$	(b) $y=5x$	(c) $f(x)=-\frac{1}{3}x$
(d) $f(x)=\frac{x}{6}$	(e) $y=x$	(f) $f(x)=x^2$
(g) $f(x)=x^3$	(h) $y=\frac{2}{3}x^6$	(i) $f(x)=7x^5$
(j) $f(x)=\frac{2x^2}{5}$	(k) $y=-x^4$	(l) $f(x)=\frac{1}{2}x^3$
(m) $y=\frac{2}{3}x^{-6}$	(n) $y=-2x^{-4}$	(o) $f(x)=5x^{-3}$
(p) $f(x)=x^{-2}$	(q) $y=\frac{-3}{x^5}$	(r) $y=\frac{-2}{3x^4}$

2. Find the gradient function for each of the following, writing your answer in positive index form and in surd form: *[power is a fraction]*

(a) $y=x^{\frac{1}{4}}$	(b) $f(x)=2x^{0.4}$	(c) $y=-3x^{\frac{2}{3}}$
(d) $y=\frac{x^{\frac{1}{4}}}{4}$	(e) $y=8x^{\frac{3}{4}}$	(f) $y=\frac{-2x^{\frac{2}{3}}}{3}$
(g) $y=\sqrt{x}$	(h) $y=4\sqrt{x^3}$	(i) $f(x)=-7\sqrt[4]{x}$
(j) $y=\frac{-2}{3x^{0.25}}$	(k) $y=-2x^{-\frac{3}{2}}$	(l) $y=\frac{1}{x}$
(m) $f(x)=\frac{1}{\sqrt{x}}$	(n) $y=\frac{-2}{3\sqrt{x}}$	(o) $f(x)=\frac{1}{5\sqrt{x^3}}$
(p) $y=x\sqrt{x}$	(q) $f(x)=\frac{\sqrt{x}}{x}$	(r) $y=\frac{x^2}{\sqrt{x}}$

Key Idea – Derivative of Powers of *x* using Rule 1 and 2

1. Find the derivative of each of the following, leaving your answer in positive index form.

(a) $y=1+3x^4$	(b) $y=x^2-3x+1$	(c) $f(x)=2x^{-3}+\frac{1}{2}$
(d) $y=2x+\frac{x^4}{2}-5$	(e) $f(x)=\frac{x^2}{4}+3x-12.5$	(f) $y=\frac{x^3}{6}+\frac{2x}{3}$
(g) $y=2x^3-\sqrt[4]{x}+\frac{2}{x^2}+1$	(h) $y=\sqrt[5]{x}+x^{-3/4}$	(i) $y=3x+\frac{3x^2}{\sqrt{x}}$

2. Differentiate the following by first expand or simplifying:

(a) $y = x(2x+1)$ (b) $y = (1+x^2)(x^2+3x)$ (c) $y = \frac{1}{5}\left(7x^3 - \frac{1}{2}x^2 + 2x - 3\right)$

(d) $f(x) = (x+4)(x-4)$ (e) $f(x) = (3x-1)^2$ (f) $f(x) = (1+x^3)^2$

(g) $g(x) = \dfrac{2x^2+3x^3}{x^2}$ (h) $f(x) = \dfrac{8x^2+6x+1}{2x}$ (i) $y = \dfrac{2x^3+5x}{x}$

<u>Week 3B</u>

Key Ideas – Basic Application of Derivatives

1. If $f(x) = x^2 + 6x$, find the value of x for which $f'(x) = 0$
2. Find the gradient of the curve $y = x^2 - 1$ when $x = 3$.
3. Find the coordinates of the point on the curve $y = x^2 - 1$, where the gradient is −8.
4. A function $y = 15x - 2x^3$ has a tangent to the curve at the point (1, 13). Find its gradient.
5. Find the gradient of the function $f(x) = 3x^2 - 2x + 5$ at .
6. Find the coordinates of the point on the curve $y = 2x^2 + 3x$ where the gradient is equal to 15.
7. A function is given by $f(x) = \dfrac{1}{x}$. Evaluate $f'(2)$.
8. If $f(x) = 12x - x^3$, find all values of x for which $f'(x) = 0$.

Key Idea – Higher Order Derivatives

1. Find the first 4 derivatives of $f(x) = x^7 - 2x^5 + x^4 - x - 3$.
2. Find $\dfrac{d^2y}{dx^2}$, if $y = x^3 - \dfrac{5}{x}$
3. Find $f''(-1)$, if $f(x) = x^4 - x^3 + 2x^2 - 5x - 1$.
4. Find $f''(-2)$, if $f(x) = x^4 - 2x + 1$.
5. For $y = x^5 - 3x^4 + x$, find $\dfrac{d^5y}{dx^5}$. 6. If $g(x) = \sqrt{x}$, find $g''(4)$
7. For $g(x) = 6x^5 + 2x^4 - 4x^3 + 7x^2 - 8x + 3$, evaluate $g^{(7)}(x)$.

Key Idea – Rule 3: Product Rule

1. Find the derivative of each of the following.

   (a) $f(x) = (x-2)(6x+7)$ (b) $y = x^3(2x+3)$ (c) $f(x) = (3x+1)(5-x)$

   (d) $y = (3x^2-1)(1-2x)$ (e) $y = (x^2-5x)(2x+3)$ (f) $f(x) = -3x^2(5x^4-x)$

   (g) $y = (5x-8)(2x+3)$ (h) $y = (x^4-1)(1-x^2)$ (i) $f(x) = (x^3+1)(x^2-3x)$
2. Given $f(x) = 2x(1+x)^3$ find $f'(x)$ and $f'(0)$.
3. Find the gradient of the tangent to the curve $y = 3x^2(2x-1)$ at $x = 2$.
4. Find the gradient of the tangent to the curve $f(x) = \sqrt{x}(x^2-2)$ at $x = 2$.
5. If $f(x) = (5x-3)(2x^3-x)$, determine the value of $f'(1)$.
6. Find the exact value of x for which the gradient of the tangent to the curve $f(x) = 2x(x^2+6x+9)$ is 14.
7. At what point on the curve $y = (2x+3)(x-1)$ is the gradient −3?

Week 3C

Key Idea – Quotient Rule

Differentiate each of the following using the quotient rule:

1. (a) $f(x)=\dfrac{3x}{x+5}$ (b) $y=\dfrac{x-2}{5x+7}$ (c) $f(x)=\dfrac{x-5}{x^2}$

   (d) $y=\dfrac{x-5}{2x+3}$ (e) $y=\dfrac{1+x^2}{1-x^2}$ (f) $f(x)=\dfrac{5x^2-2x}{2x+5}$

2. Find the gradient of the function $y=\dfrac{x^2}{x-3}$ at the point $\left(5,\dfrac{25}{2}\right)$.

3. If $f(x)=\dfrac{4x+5}{2x-1}$, evaluate $f'(2)$.

4. Find the values of x for which the gradient of the tangent to $f(x)=\dfrac{4x-1}{2x-1}$ is –2.

5. Given $f(x)=\dfrac{2x}{x+3}$, find x if $f'(x)=\dfrac{1}{6}$.

Key Idea – Rule 5: Chain Rule

1. Use the chain rule to differentiate, leaving your answer in positive index form.

   (a) $f(x)=(2x+1)^3$ (b) $y=(x^3-1)^5$ (c) $f(x)=(1-x)^4$

   (d) $y=(x^3-2x^2+5)^3$ (e) $y=3(7x+4)^7$ (f) $f(x)=\dfrac{1}{3}(5x+1)^6$

   (g) $f(x)=(2-3x)^{-4}$ (h) $y=\dfrac{2}{(x^3+3)^2}$ (i) $f(x)=\dfrac{-5}{(x^3-7x^2+x)^2}$

   (j) $f(x)=\sqrt{5x-3}$ (k) $y=\dfrac{6}{\sqrt{4+x}}$ (l) $y=\dfrac{2}{(4x-1)^{\frac{1}{5}}}$

2. Given $f(x)=\dfrac{(x^2+1)^4}{2}$ find $f'(1)$.

3. Given $f(x)=3+4(2x-1)^3$ solve $f'(x)=0$

4. For the function $f(x)=5(1-x)^3$ evaluate $f'(-1)$.

Key Idea – Derivative of Powers of *x* Combining Rule 1-5

1. Differentiate the following, leaving your answer in surd form if required.

   (a) $f(x)=2x^5(5x+3)^3$ (b) $y=(3x-4)\sqrt{5-2x}$

2. Find the gradient of the tangent to the curve $y=\dfrac{\sqrt{4-x^2}}{x+1}$ at the point where $x=0$.

3. Find the derivatives of the following. Taken from MATH141

   (a) $y=\dfrac{x^3-17}{(x^4-3)^5}$ (b) $y=\dfrac{x^3-17}{(x^4-3)^5}$

Week 3D

Key Idea – Further Applications of Derivatives - Equation of Tangent

1. For each of the following, write the equations in the general form.
   (a) Find the equation of the tangent to the curve $y = x^2$ at (2, 4).
   (b) Find the equation of the tangent to the parabola $y = x^2 - 5x + 6$ where the parabola cuts the y-axis.
   (c) Find the equation of the tangent to the curve $y = 3x^3 - 7x^2 + 2x$ where $x = 2$.
   (d) Find the equation of the tangent to the parabola $y = 2x^2 - 4x + 1$ where the gradient is 4.
   (e) Find the equations of the tangents to the parabola $y = x^2 - x - 12$ where the parabola cuts the x-axis.
   (f) Find the equation of the tangent to the curve $y = 3x^2 - 8x + 6$ at the point where the tangent is parallel to the line $4x - y + 20 = 0$.

Key Idea – Derivatives of Exponential Functions

1. Differentiate the following:

(a) $y = e^{4x}$ (b) $y = e^{-x}$ (c) $y = e^{4x+1}$

(d) $y = -e^{x^2-4x+1}$ (e) $y = 2e^{3x^4}$ (f) $y = \dfrac{e^{6x}}{2}$

(g) $y = 2x^3 + e^{5x}$ (h) $y = 3x^2 + e^{1-x} + 3$ (i) $f(x) = e^{2x}\left(e^x - e^{-x}\right)$

2. Differentiate the following:

(a) $y = 2x^3e^{5x}$ (b) $y = \dfrac{e^{2x}+1}{2x-1}$ (c) $f(x) = 3\left(2e^x - 1\right)^6$

(d) $f(x) = \left(x + e^{2x}\right)^3$ (e) $f(x) = x^3e^{2x-1}$ (f) $y = \dfrac{e^{2x+1}}{2x+5}$

3. (a) Given $g(x) = 4e^{2x-1}$ find $g'(2)$.
   (b) If $y = e^{2x} + e^{8x}$ find $\dfrac{d^2y}{dx^2}$.
   (c) Find the equation of the tangent to the curve $f(x) = -2e^{3x}$ when $x = 1$. Write the equation in the general form.
   (d) Find the point on the curve $f(x) = 2xe^x$ where the gradient is zero.

4. Find the equation of the tangent to $y = e^{x^3+1}$ at the point $x = -1$.